

Ethan Lo

elo9908@gmail.com • (470) 977-9692 • linkedin.com/in/ethan-lo-2307252ba • ethan-lo.com • github.com/ethanol-lo

EDUCATION

Georgia Institute of Technology, Atlanta, Georgia

May 2028

Bachelor of Science in Computer Science, Threads: AI and Modeling & Simulation

GPA: 3.77/4.00

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization & Programming, Startup Lab, Calculus I & II, Linear Algebra, Multivariable Calculus, Applied Combinatorics, Differential Equations

TECHNICAL SKILLS

ML/AI – Proficient in XGBoost, GradientBoost, Logistic Regression, KNN, RF, MLP, SVM, CNN, RNN, LSTM, BERT

Languages & Frameworks – Proficient in Python, Java; Intermediate in JUnit, HTML/CSS, JavaScript

Databases & Tools – Proficient in VS Code, IntelliJ, Git, Google Colab; Intermediate in Jupyter Notebook, Raspberry Pi

WORK EXPERIENCE

ScoutOut via Harvard Ventures Technology Summer Program, Remote – AI/ML Intern

Jun 2025 – Jul 2025

- Developed a regression model trained on 50,000 properties to predict real estate value from location and market features
- Predicted with a MAE of 0.0273 and R^2 score of 0.941 using XGBoost
- Presented interactive UI demo and analytical results to company founder (ex-Amazon)

RESEARCH PROJECTS

Generation of Folksong Melodies with Deep Learning and Valence-Arousal Emotions

Aug 2025 – April 2026

- Proposed a generative LSTM model trained on the Essen Folksong Database to create MIDI sequences of melodies
- Collected participants' emotional perception of generated melodies via online anonymous surveys
- Achieved a validation loss of 3.99; melodies demonstrated pitch/rhythmic diversity and convincing human character

Cyclone Trajectory Prediction and Status Classification with ML/DL Techniques

Sep 2025 – Nov 2025

- Applied GradientBoost, RF, SVM, and MLP techniques to forecast cyclone trajectories and classify storm intensity statuses
- Optimized algorithms for time-series weather data using sliding windows; best accuracy achieved: 93%
- Managed data preprocessing, model build, and manuscript composition (arXiv: <https://doi.org/10.48550/arXiv.2509.24146>)

Exoplanet Detection ML Model Trained on Synthetic Light Curves

Jul 2024 – Sep 2024

- Conducted research on NASA Kepler telescope data of 5,000+ exoplanet candidates and synthetic light curves using SMOTE
- Directed a 5-person team for data preprocessing and model design using logistic regression, KNN, and RF
- Achieved 92% classification accuracy; independent manuscript published (arXiv: <https://doi.org/10.48550/arXiv.2507.19520>)

PROGRAMMING PROJECTS

Mental Health Sentiment Analysis Platform

April 2025

- Collaborated with a 4-member team to build an NLP-based mental health sentiment analysis platform using BERT
- Created a React website to demonstrate functionality with data visualization tools for emotional awareness
- Earned 1st Place at the Empower Atlanta Divergent Teams Hackathon at Microsoft among 13 Georgia teams

Object Detection Using a Resource-Limited Device with Machine Learning

Jan 2024

- Built an object detection system backed by YOLOv5 supporting both image and video stream inputs
- Deployed on a Raspberry Pi with low computational overhead, achieving 20 FPS
- Received 3rd Place at the North Cobb Technology Competition in the programming category

LEADERSHIP AND COMMUNITY ENGAGEMENT

Volunteer, Overseas Community Affairs Council of Taiwan English Teaching Volunteer Service Program

July 2026

- Completed 40-hour teacher training at Taiwan Medical University
- Taught English to 11 elementary students at Fengtian Elementary School in Zhubei City for 3 weeks

Vice President, Walton High School Computer Science Honor Society

Aug 2025 – May 2026

- Established coding tutoring programs at local elementary schools
- Promoted CS education and organized hackathons and seminars at Walton

Founder, Walton Public Speaking Club

Aug 2025 – May 2026

- Empowered 20+ students in oral excellence and effective communication at biweekly meetings
- Organized the first TEDxWaltonHigh event with 5 speakers for 50+ attendees

ADDITIONAL SKILLS AND INTERESTS

Languages: Fluent in English; Conversational in Mandarin, Latin, Spanish

Fine Arts: Cello, piano, Canva design

Interests: Tennis, volleyball, pickleball, cardistry, calisthenics